ECE 456 Problem Set 1:
The Basics of Electron Flow in Nanodevices

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1 Current with a Lorentzian Density of States

1.1 Alternative Form

Given the following expressions for the number of channel electrons and the current in a nanodevice:

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E) + \gamma_2 f_2(E)}{\gamma_1 + \gamma_2} D(E-U)\,\mathrm{d}E \tag{1}$$

$$I = \frac{q}{\hbar}\frac{\gamma_1\gamma_2}{\gamma_1+\gamma_2}\int_{-\infty}^{\infty} [f_1(E) - f_2(E)] D(E-U)\,\mathrm{d}E \tag{2}$$

These expressions can be rewritten into a form that is more convenient for numerical implementation in MATLAB. First, define a change of variables:

$$\begin{aligned} E' &= E - U \\ E &= E' + U \\ \mathrm{d}E' &= \mathrm{d}E \end{aligned}$$

Since subtracting a constant does not alter the limits of integration, the integration bounds remain:

$$E' : -\infty \to \infty$$

Substituting $E = E' + U$ and relabeling $E' \to E$ yields:

$$N = \int_{-\infty}^{\infty} \frac{\gamma_1 f_1(E+U) + \gamma_2 f_2(E+U)}{\gamma_1 + \gamma_2} D(E)\,\mathrm{d}E \tag{3}$$

$$I = \frac{q}{\hbar}\frac{\gamma_1\gamma_2}{\gamma_1+\gamma_2}\int_{-\infty}^{\infty} [f_1(E+U) - f_2(E+U)] D(E)\,\mathrm{d}E \tag{4}$$

1.2 Solving the Equations Self-Consistently

Using the code provided in `problem1.py`, the values of N and I were computed self-consistently. The resulting plots are shown in Figure 1.

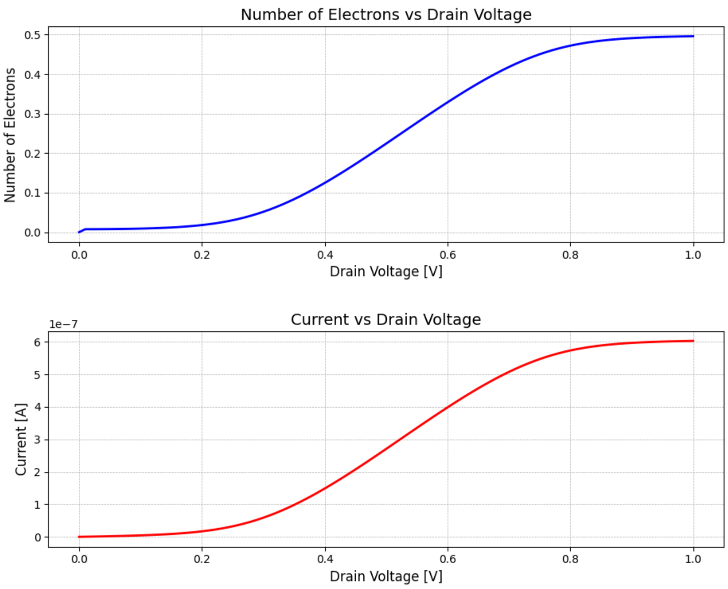

Figure 1: Current (I) and number of electrons (N) as functions of drain voltage (V_D).

1.3 Difference Proportional to Current

By plotting the difference between the Fermi functions using a discrete energy level (modeled as an impulse) rather than a broadened Lorentzian density of states, the relationship between the current I and the difference in the source and drain Fermi functions becomes clearer.

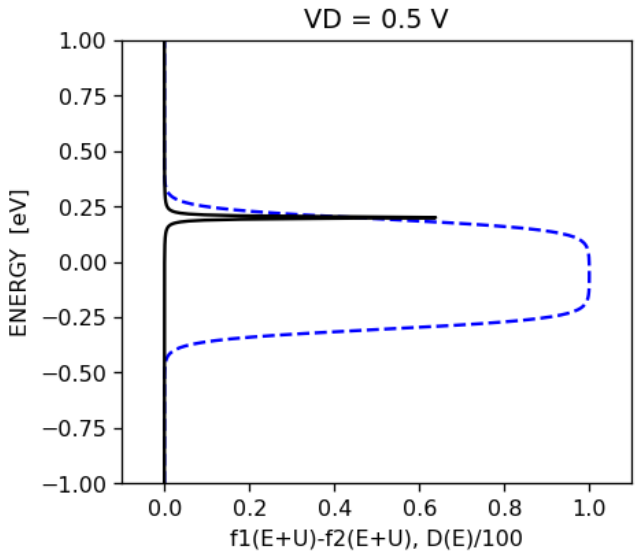

Figure 2: Difference between the Fermi functions as a function of energy, with the density of states modeled as an impulse.

V_D (V)	$[f_1(E+U) - f_2(E+U)]_{E=\varepsilon}$	I (A)
0.2	0.015	1.69×10^{-8}
0.5	0.440	2.705×10^{-7}
0.8	0.970	5.736×10^{-7}
1.0	1.000	6.030×10^{-7}

Table 1: Comparison of the Fermi-function difference and current values at identical drain voltages, evaluated at $\varepsilon = 0.2\,\text{eV}$.

The ratio of $[f_1(E+U) - f_2(E+U)]_{E=\varepsilon}$ at $V_D = 1.0$ to that at $V_D = 0.8$, and the ratio at $V_D = 0.8$ to that at $V_D = 0.5$, are found to be 1.031 and 2.205, respectively.
Similarly, the corresponding current ratios are 1.051 and 2.12. These results confirm that the current scales proportionally with the difference in the Fermi functions.

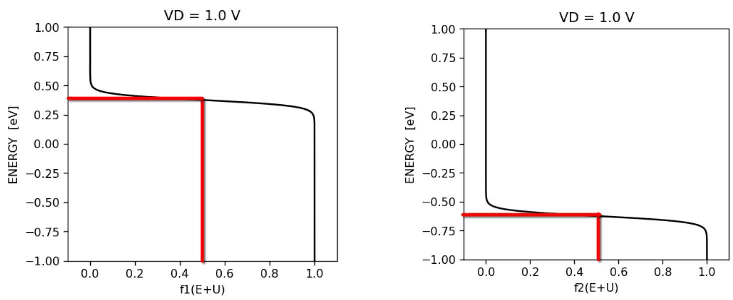

Figure 3: Step locations of the Fermi functions at $V_D = 1.0\,\text{V}$.

At $V_D = 1.0\,\text{V}$, the step locations of the Fermi functions were determined to be $\mu_1 = 0.3760$ and $\mu_2 = 0.6240$. Using these values in `problem1c3.py` yields a self-consistent potential of $U = -0.250\,27\,\text{eV}$.

At this drain voltage, the source contact (f_1) favors electron injection into the channel, while the drain contact (f_2) favors electron absorbtion from the channel. This can be visualized in Figure 3 where the Fermi function of f_1 steps down at a higher energy than f_2. These two biases produce current flow through the channel. Therefore, energies that ε can be located must be less than $\mu_1 = 0.3760$ and greater than $\mu_2 = 0.6240$.

When $V_D = 0$, the electrochemical potentials of the source and drain are equal ($\mu_1 = \mu_2$), which implies that their Fermi functions are identical ($f_1(E) = f_2(E)$). Consequently, the source is trying to fill (or empty) the channel level to the exact same extent that the drain is. The lack of a difference in agenda leads to no net movement of electrons, and thus the current is zero.

2 Current in a Nanoscale MOSFET